EE 541 HW 6

1) $W^{(1)} = \begin{bmatrix} 1 & -2 & 1 \\ 3 & 4 & -2 \end{bmatrix}$, $b^{(1)} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$

$W^{(2)} = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$, $b^{(2)} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$W^{(3)} = \begin{bmatrix} 2 & 2 \\ 3 & -3 \\ 2 & 1 \end{bmatrix}$, $b^{(3)} = \begin{bmatrix} 0 \\ -4 \\ -2 \end{bmatrix}$

a) Feedforward table

$x = [+1 \quad -1 \quad +1]^T$

Layer 1

$s^{(1)} = W^{(1)} x + b^{(1)}$

$\Rightarrow [4, -3] + [1 \; -2] = [5 \; -5]$ //

$\Rightarrow a^{(1)} = \text{ReLU} = [5, 0]$ //

$\Rightarrow a'^{(1)} = [1, 0]$ //

Layer 2:

$s^{(2)} = W^{(2)} a^{(1)} + b^{(2)} \Rightarrow [5, 15] + [1, 0]$

$= [6, 15]$ //

$a^{(2)} = \text{ReLU} = [6, 15]$

$a'^{(2)} = [1, 1]$ //

Layer 3

$$s^{(3)} = W^{(3)} a^{(2)} + b^{(3)}$$

$$\Rightarrow \begin{bmatrix} 2 & 2 \\ 3 & -3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 6 \\ 15 \end{bmatrix} + \begin{bmatrix} 0 \\ -4 \\ -2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 42 \\ -27 \\ 27 \end{bmatrix} + \begin{bmatrix} 0 \\ -4 \\ -2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 42 \\ -31 \\ 25 \end{bmatrix}$$

$$a^{(3)} = \text{softmax}([42, -31, 25])$$

$$\Rightarrow \frac{[e^{0}, e^{-73}, e^{-17}]^T}{e^{0} + e^{-73} + e^{-17}} = \begin{bmatrix} 1 \\ 2\times10^{-32} \\ 4.1\times10^{-8} \end{bmatrix} \approx \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \rightarrow \text{using this for below}$$

b)

$$\delta^{(3)} = a^{(3)} - y = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$\delta^{(2)} \Rightarrow W^{(3)T} \delta^{(3)} = \begin{bmatrix} 2 & 2 \\ 3 & -3 \\ 2 & 1 \end{bmatrix}^T \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$\delta^{(1)}$:

$$W^{(2)T} \cdot \delta^{(2)} = \begin{bmatrix} 1 & 3 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$\delta^{(1)} = \dot{a}^{(1)} \cdot \begin{bmatrix} 3 \\ 4 \end{bmatrix} \Rightarrow \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

$$W_{new} = W - \eta \cdot \delta \cdot (a_prev)^T$$

$$b_{new} = b - \eta \cdot \delta$$

<u>Layer 1</u>

$$a_prev = x = [1, -1, 1]^T, \quad \delta^{(1)} = [3, 0]^T$$

$$\delta^{(1)} \cdot x^T \Rightarrow \begin{bmatrix} 3 \\ 0 \end{bmatrix} [1, -1, 1] = \begin{bmatrix} 3, & -3, & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

$$W^{(1)}_new = \begin{bmatrix} 1 & -2 & 1 \\ 3 & 4 & -2 \end{bmatrix} - 0.5 \begin{bmatrix} 3 & -3 & 3 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} -0.5 & -0.5 & -0.5 \\ 3 & 4 & -2 \end{bmatrix}$$

$$b^{(1)}_new = [1, -2] - 0.5[3, 0] = [-0.5, -2]^T$$

<u>Layer 2:</u>

$$a_prev = a^{(1)} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}, \quad \delta^{(2)} = [0, 1]^T$$

$$\delta^{(2)} \cdot a^{(1)T} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} (5, 0) = \begin{bmatrix} 0 & 0 \\ 5 & 0 \end{bmatrix}$$

$$W^{(2)}_new = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix} - 0.5 \begin{bmatrix} 0 & 0 \\ 5 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 0.5 & 4 \end{bmatrix}$$

$$b^{(2)}_new = [1, 0] - 0.5[0, 1] \Rightarrow [1, -0.5]$$

Layer 3 :

$a\text{-prev} = a^{(2)} = [6, 15]^T, \quad \delta^{(3)} = [1, 0, -1]^T$

$$\delta^{(3)} \cdot a^{(2)} \Rightarrow \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} [6, 15] \Rightarrow \begin{bmatrix} 6 & 15 \\ 0 & 0 \\ -6 & -15 \end{bmatrix}$$

$$W^{(3)}\text{-new} \Rightarrow \begin{bmatrix} 2 & 2 \\ 3 & -3 \\ 2 & 1 \end{bmatrix} - 0.5 \begin{bmatrix} 6 & 15 \\ 0 & 0 \\ -6 & -15 \end{bmatrix} \Rightarrow \begin{bmatrix} -1 & -5.5 \\ 3 & -3 \\ 5 & 8.5 \end{bmatrix}$$

$$b^{(3)}\text{-new} \Rightarrow [0, -4, -2] - 0.5 \cdot [1, 0, -1] = [-0.5, -4, -1.5]$$